

RETAILPAY TEST

Background

A client, **KK Wholesalers**, operates a distributed wholesale business consisting of:

- **2 Branches**
- Branch A → **1 Store**
- Branch B → **2 Stores**

Each store independently performs sales while stock movement may occur:

- Between stores within a branch
- Between branches
- From centralized procurement

The business has experienced **stock inconsistencies**, **race conditions during peak sales**, and **audit challenges**.

RetailPay has contracted you to design and implement a **production-grade inventory movement system** capable of supporting future scale.

The company currently manages **10 SKUs**, but expects growth to **100,000+ SKUs and high transaction volume**.

Objective

Design and implement a system that models:

- Sales
- Inter-store transfers
- Inter-branch transfers
- Stock adjustments
- Complete stock movement traceability
- Roles (Minimal Implementation) . Model only Administrator, Branch Manager, Store Manager

Authorization logic should demonstrate extensibility.

The system must guarantee:

- Stock accuracy
- Transaction integrity

- Auditability
- Scalability

Required Stack

- **Laravel (latest stable version)**
- MySQL / MariaDB
- VueJS OR Blade

Evaluation Criteria

Area	What We Assess
Architecture	Senior system thinking
Data Modeling	Correct domain abstraction
Concurrency Handling	Real-world readiness
Laravel Mastery	Framework depth
Database Design	Performance awareness
Code Quality	Maintainability
Communication	Engineering reasoning
Testing Strategy	Risk awareness

NB

Please note that this assignment is purely a skill assessment test. Though related to our line of work, the assignment does not form part of any future or current task descriptions.

Good luck.